

Transform the following statements into their logical equivalents after negation.

1. $\neg(\forall x(P(x) \rightarrow Q(x)))$

Solution: $\exists x(P(x) \wedge \neg Q(x))$

2. $\neg(\exists x(P(x) \wedge Q(x)))$

Solution: $\forall x(\neg P(x) \vee \neg Q(x))$

3. $\neg(\forall x \exists y R(x, y))$

Solution: $\exists x \forall y \neg R(x, y)$

Translate
English $\rightarrow$ Logic

1- "Every student has a laptop"

$S(x)$: x is a student

$L(x)$: x has a laptop

Solution: $\forall x (S(x) \rightarrow L(x))$

2- "Every student taking a computer science course owns a laptop"

$S(x)$: x is a student

$C(x)$: x is taking a computer science course

$P(x)$: x owns a laptop

Solution: $\forall x ((S(x) \wedge C(x)) \rightarrow P(x))$

3- "Some students owns a laptop"

$S(x)$: x is a student

$C(x)$: x is taking a computer science course

$P(x)$: x owns a laptop

Solution: $\exists x (S(x) \wedge P(x))$

4- "No student owns a laptop"

$S(x)$: x is a student

$P(x)$: x owns a laptop

Solution: $\neg \exists x (S(x) \rightarrow P(x))$

5- "Every student has at least one friend"

$S(x)$: x is a student

$P(x)$: x owns a laptop

$F(x, y)$: x is friend with y

Solution: ~~_____~~

$\forall x (S(x) \rightarrow \exists y F(x, y))$

6- "Everyone likes someone"

$L(x,y)$: x likes y

Solution: $\forall x \exists y L(x,y)$

7- "Someone likes everyone"

$L(x,y)$: x likes y

Solution: $\exists x \forall y L(x,y)$

8- "Everyone has exactly one best friend"

$F(x,y)$: x is a friend of y , $B(x,y)$: x is y 's best friend

Solution: $\forall x \exists! y B(x,y)$